



IIC Kwajalein Geoid:
Development of a Geoid for Kwajalein
Atoll, Republic of Marshall Islands

Assignment Title:

**LiDAR Survey Data Quality Assurance
and Quality Control**

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Prepared By: Dr David Dodd

IIC Technologies Ltd
4 Palawan Place
Torbay
Auckland
New Zealand

E-mail info@iictechnologies.com
Website www.iictechnologies.com

Abstract

Background.

IIC Technologies (IIC), under contract to The Pacific Community (SPC), was contracted to provide Quality Control and Quality Assurance of contractor provided topographic and bathymetric LiDAR Data of the Republic of the Marshall Islands (RMI). As a follow on to this work, SPC requested IIC conduct a review into the existing geodetic and height data and surveys to determine the cause of vertical discrepancies noted in the RMI geodetic information. The results of the initial review determined and reported the primary cause for the 1m vertical discrepancy between the LiDAR Contractor data with the geodetic control at Majuro and made recommendations on how these should be mitigated (see *IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*). In addition to the 1m error, further discrepancies were noted in that report. Therefore, SPC contracted IIC to conduct analysis and reporting on the remaining height discrepancies and generate a new GEOID for Majuro (see *IIC Majuro Geoid: Development of a Geoid for Majuro Atoll, Republic of Marshall Islands dated 26 May 2020*).

With the issues at Majuro resolved and a new GEOID established, for consistency SPC requested that IIC undertake a final body of work to review the geodetic and survey information and generate a geoid model for the Kwajalein Atoll, which including Ebeye. This report highlights the findings of that review and information on the creation of this new geoid.

Summary.

A new geoid was created for the entire Kwajalein Atoll, including Ebeye, by vertically adjusting the EGM08 geoid. The new geoid was adjusted to MSL as defined by the NOAA 2019 survey and 7 months of tide gauge observations at Ebeye.

The final geoid provides a model to translate between ellipsoid heights, referenced to ITRF2014 epoch 2019, and orthometric heights, referenced to MSL as defined by the 2019 survey 1.468m above tide gauge zero and 1.482m below BMWHRF. This datum is referred to as “EbeyeGVD19”.

This report provides the findings of the review and information on the creation of the new geoid “KGeoid2020_ITRF14ep2019_EbeyeGVD19”.

This analysis and Report was undertaken and prepared by Dr David Dodd.



IIC's Project Manager and Point of Contact:

David Crossman
4 Palawan Place
Torbay
Auckland
0630
New Zealand
E-mail David.crossman@iictechnologies.com
Tel: +64 (0) 272 772722

E-mail info@iictechnologies.com

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1 Introduction

IIC Technologies (IIC), under contract to The Pacific Community (SPC), was contracted to provide Quality Control and Quality Assurance of contractor provided topographic and bathymetric LiDAR Data of the Republic of the Marshall Islands (RMI). As a follow on to this body of work, IIC undertook investigations into vertical discrepancies and generated a new Geoid at Majuro (see IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020 and IIC Majuro Geoid: Development of a Geoid for Majuro Atoll, Republic of Marshall Islands dated 26 May 2020). Finally, at SPC request IIC has reviewed relevant data and generated a geoid model for the Kwajalein Atoll, including Ebeye, the findings and information of which are contained in this report.

Ebeye is located approximately 500 km NW of Majuro (see Figure 1).

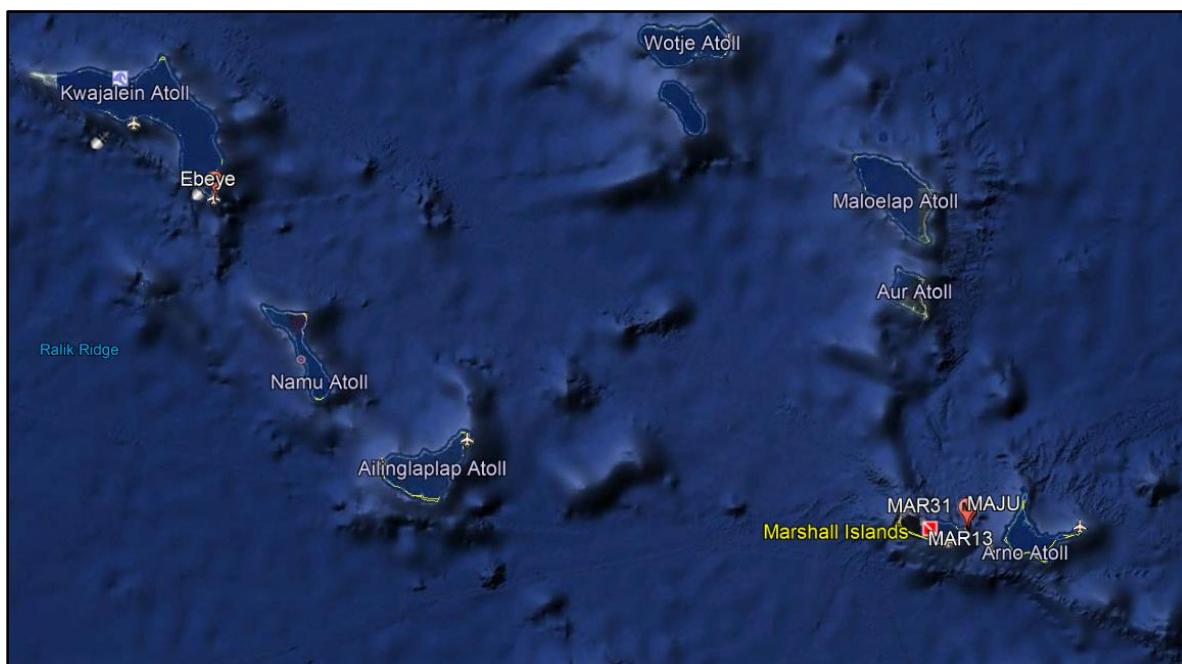


Figure 1: Kwajalein and Majuro Atolls

A new geoid was created for the entire atoll by vertically adjusting the EGM08 geoid. The new geoid was adjusted to MSL as defined by the NOAA 2019 survey, including 7 months of tide gauge observations at Ebeye.

The final geoid provides a model to translate between ellipsoid heights, referenced to ITRF2014 epoch 2019, and orthometric heights, referenced to MSL as defined by the 2019 survey 1.468m above tide gauge zero and 1.482m below BMWHRF. This datum is referred to as “EbeyeGVD19”.

This report provides the findings of the review and information on the creation and validation of the new geoid “KGeoid2020_ITRF14ep2019_EbeyeGVD19”.

2 Orthometric Datum

The orthometric datum for Ebeye was derived from a NOAA 2019 survey. A temporary tide gauge was installed from December 2018 to July 2019, being approximately 7 months. Ellipsoid heights were determined for the BM BWHRF and MSL (from 7-month average). See Figure 2.

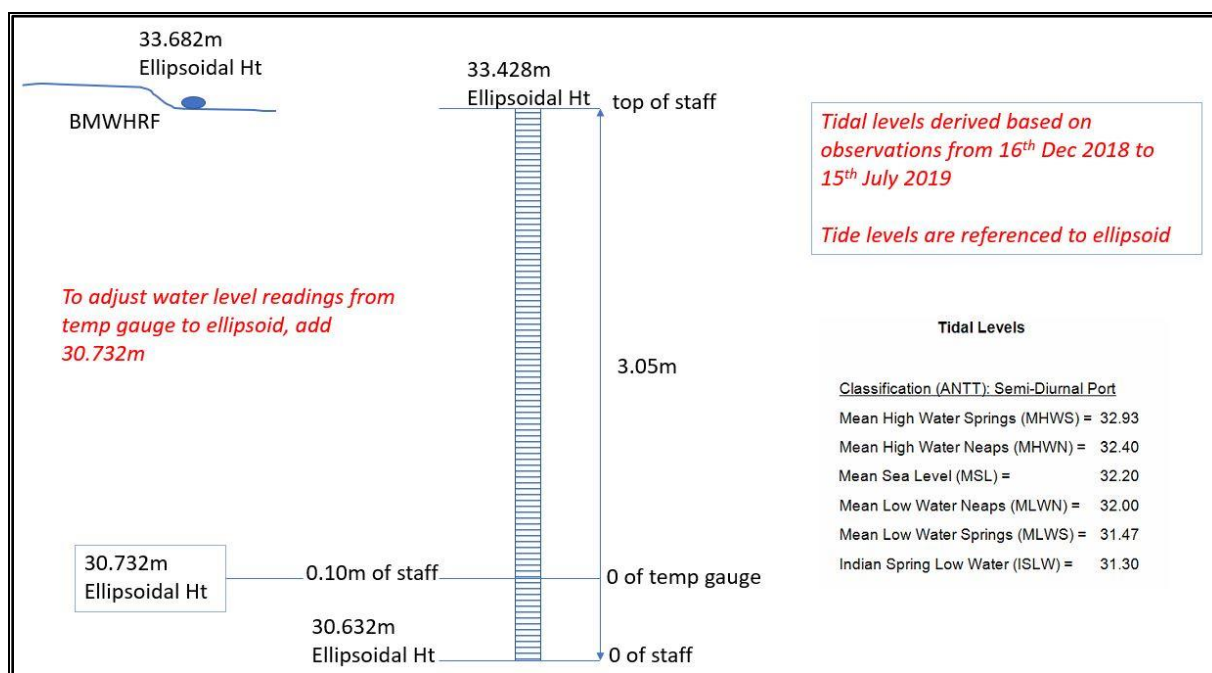


Figure 2: NOAA Ebeye temporary tide gauge datum information

Figure 3 shows the relationship between EGM08, the ITRF2019 ellipsoid, EbeyeGVD19 for tide gauge zero (TGZ) and BMWHRF.

- EGM08 to EbeyeGVD19 = 0.841m
- TGZ to EbeyeGVD19 = 1.468m

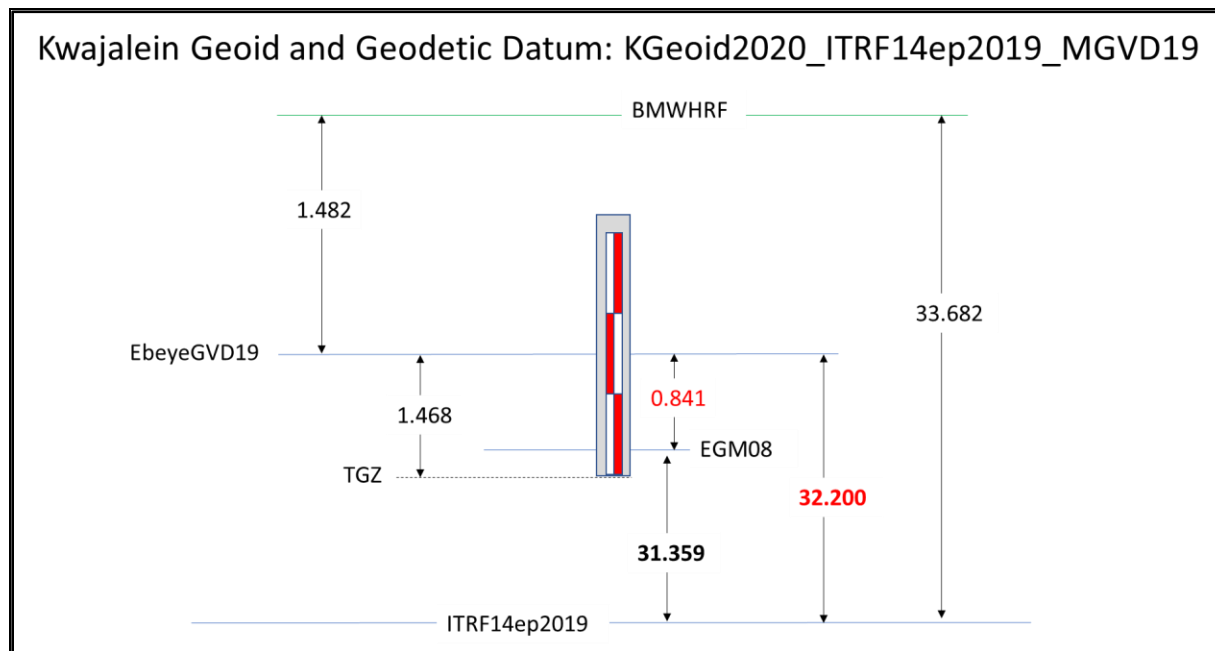


Figure 3: Ebeye Datums

The geoid model was created by first extracting a 1-minute grid of EGM08 values using the “gridget_1min.exe” function (NGA). A 0.001° (~100m) Geotiff was interpolated from the EGM08 model values. This Geotiff was shifted vertically by 0.841m to create the final “KGeoid2020_ITRF14ep2019_EbeyeGVD19” surface.

EGM08 values for a file of control points (Ebeye points.CSV), with ellipsoid heights, were interpolated using “gridget_1min.exe” and adjusted by 0.841m to create orthometric heights (see “Ebeye_points_WithEGM08_N_EbeyeGVD19.xlsx”).

No other orthometric elevations were available at the time of this report; therefore, the resulting geoid was not able to be validated. It is therefore recommended that the geoid be validated utilising field observations when possible.

3 Kwajalein Geoid

A geoid is used to represent the surface of the earth based on a surface of constant gravity. A geoid model is used to represent this surface, relative to a reference ellipsoid. In order to create a local geoid, two datums must be defined; the ellipsoid reference and the orthometric reference, where we define the surface of constant gravity. A geoid model changes spatially because of gravity variations due to local variations in the earth's mass (e.g. mountains, seamounts, density of crust material). The model also varies because of the shape of the ellipsoid. A reference ellipsoid is a mathematical representation of the earth's surface and does not vary due to gravity.

Figure 4 depicts the relationship between the ellipsoid and geoid. This relationship is defined through the following equation:

$$h = N + H$$

Where:

- h = ellipsoid height
- H = orthometric height
- N = geoid height (or geoid undulation)

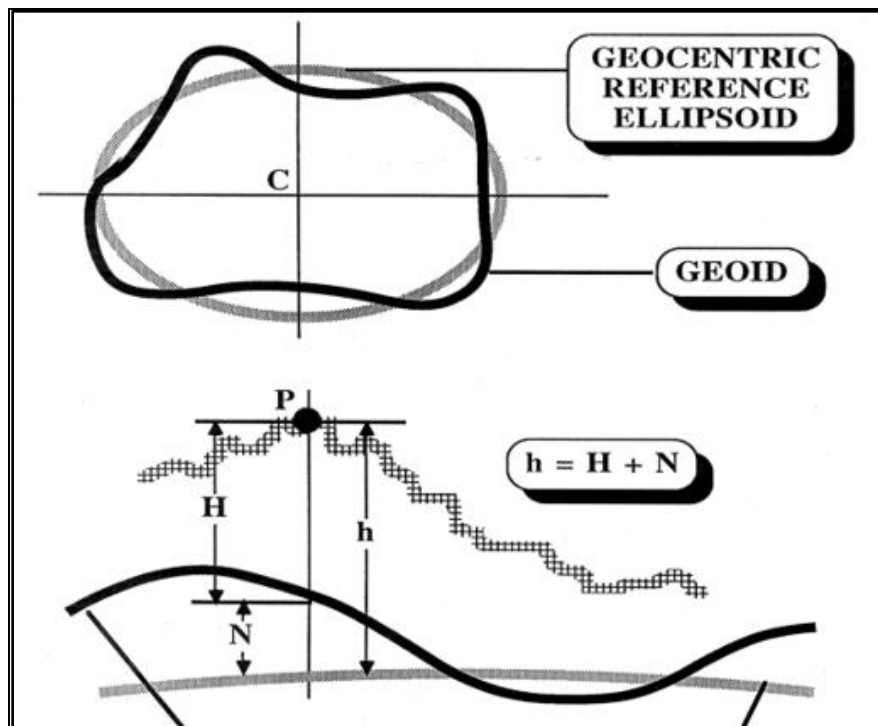


Figure 4: Geoid and ellipsoid relationship

One version of a geoid model (model of N values) is the Earth Gravity Model of 2008 (EGM08), created and distributed by the US National Geospatial-Intelligence Agency (NGA), see:

https://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08_wgs84.html.

This model was developed to best represent the geoid, globally, relative to the WGS84 reference frame (and ellipsoid).

The Kwajalein geoid model was created by first extracting a 1-minute grid of EGM08 values using the “gridget_1min.exe” function (NGA). A 0.001° (~100m) Geotiff was then interpolated from the EGM08 model values. This Geotiff was shifted vertically by 0.841m to create the final “KGeoid2020_ITRF14ep2019_EbeyeGVD19” surface (see Figure 5).

EGM08 values for a file of control points (Ebeye points.CSV), with ellipsoid heights, were interpolated using “gridget_1min.exe” and adjusted by 0.841m to create orthometric heights (see “Ebeye_points_WithEGM08_N_EbeyeGVD19.xlsx”). No other orthometric elevations were available at the time of this report; therefore, the field validation of the geoid is recommended when possible.

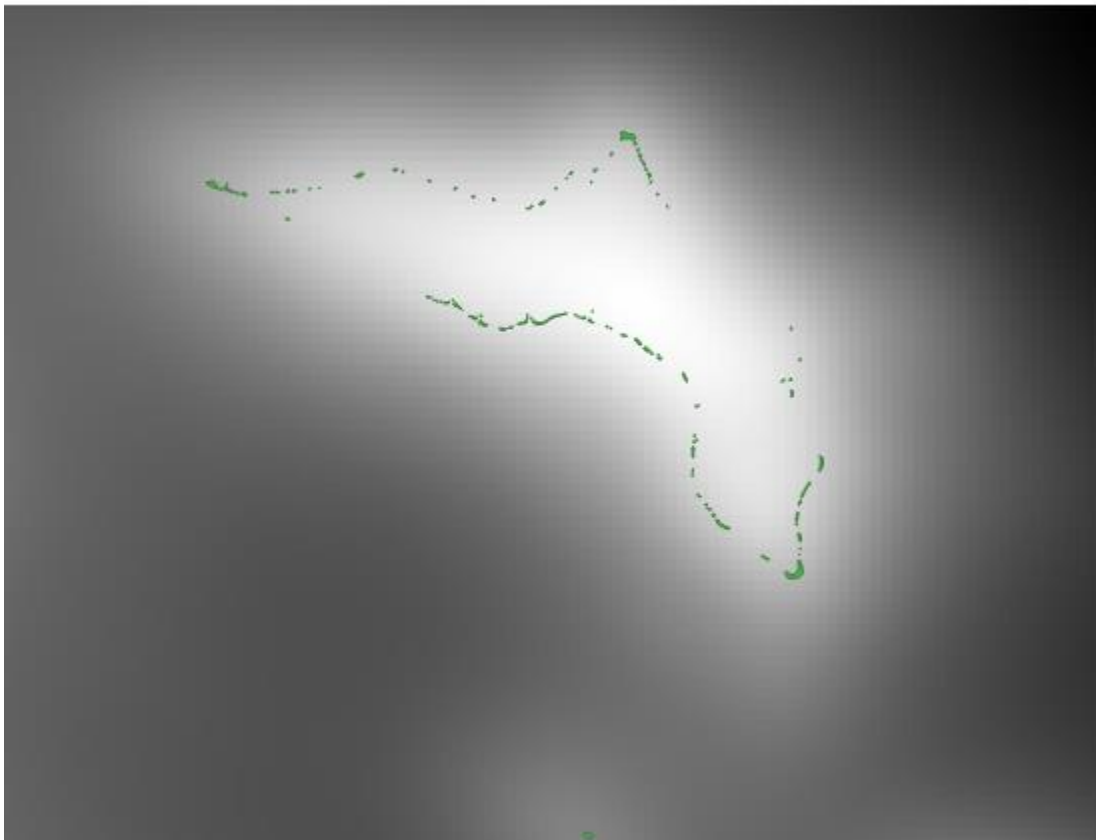


Figure 5: Kwajalein Geoid

4 Tide Gauges Comparison

A review of local tide gauge data was performed to determine the MSL (and datum) relationships. At the time of the Ebeye tide gauge observations in late 2018 and mid 2019, the Majuro gauge was operational, as was a permanent NOAA gauge on the southern part of Kwajalein. Data from all gauges were plotted and compared.

Information for the Kwajalein NOAA TG can be found at

<https://tidesandcurrents.noaa.gov/waterlevels.html?id=1820000>

- MSL = 0.567m above MLLW (CD) Epoch 1983-2001
- MSL for 16 Dec 2018 to 15 July 2019 = 0.661m above MLLW (offset of 0.094m)

Majuro TG

- MGVD17 = 1.102m above TGZ.
- MSL for 16 Dec 2018 to 15 July 2019 = 0.035m above MGVD17.

All three datasets are plotted in Figure 6, Figure 7 and Figure 8, showing all of April, one April spring tide and one April neap tide. The tide data was adjusted to the datum at their respective locations. The plots clearly indicate that the tidal signals were very similar.

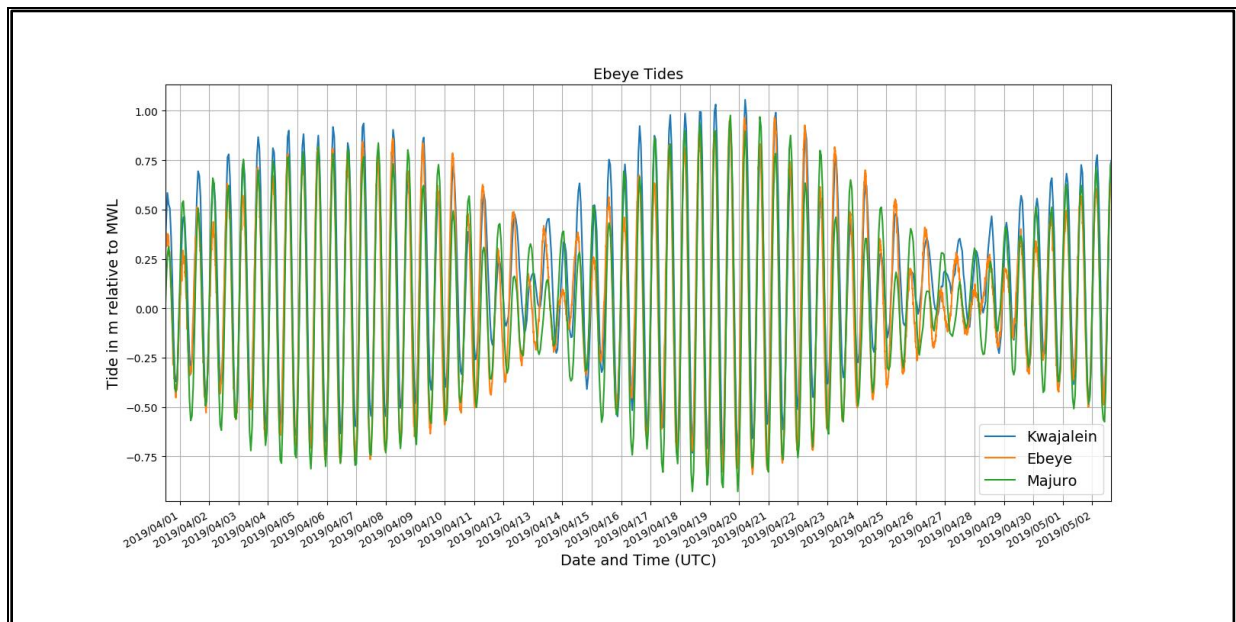


Figure 6: Tides for Majuro, Ebeye and Kwajalein, April 2019

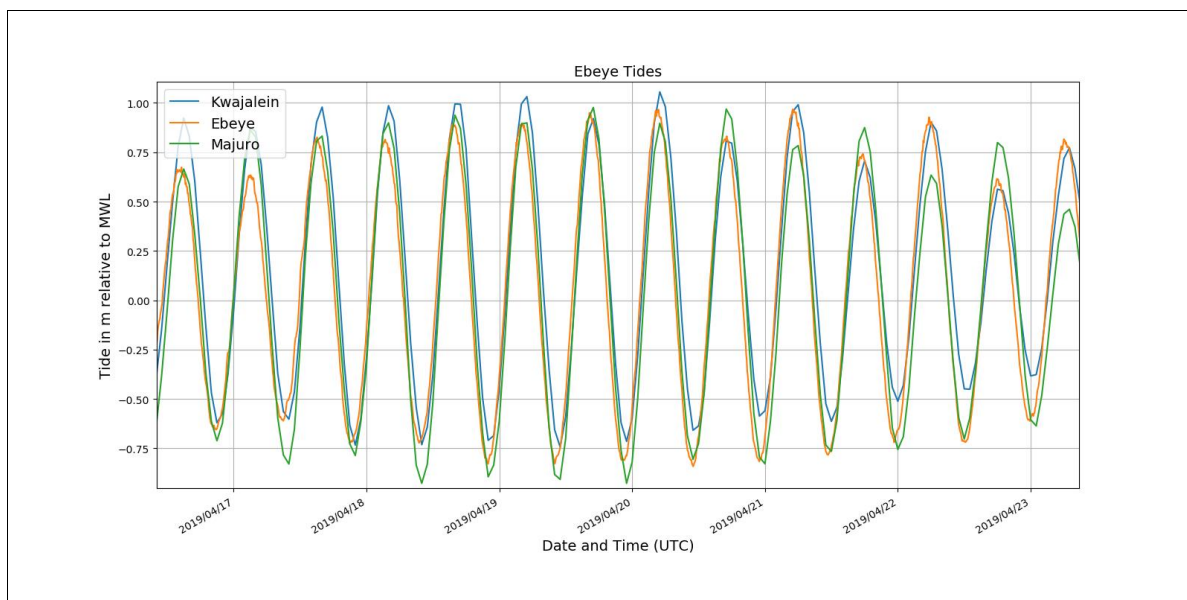


Figure 7: Tides for Majuro, Ebeye and Kwajalein, April spring tide, 2019

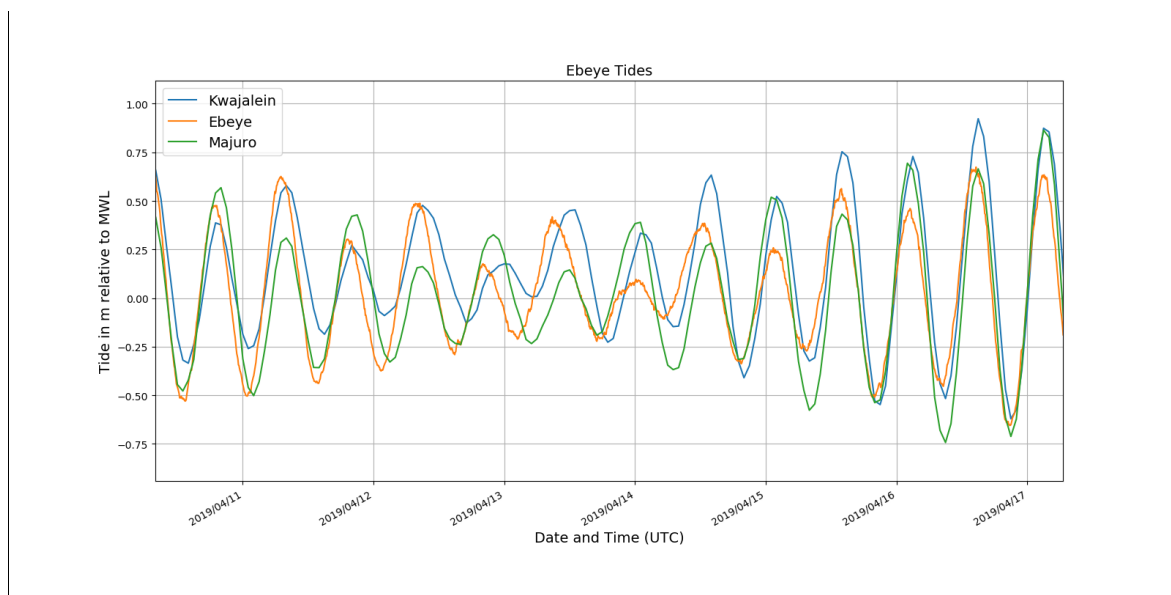


Figure 8; Tides for Majuro, Ebeye and Kwajalein, April neap tide, 2019

The relationship between the tide observations is shown in Table 1.

The mean value indicates the difference between the established MSL datum and the MSL derived from the 7 months of observations.

Ebeye and Majuro are very close (within a few CM) because their datums were derived from more recent observations than Kwajalein, which was from epoch 1983 to 2001. The standard deviation is an indicator of the tidal range. All are very similar, with Kwajalein and Ebeye being identical, and Majuro being 2.6cm different.

	Mean	Standard Deviation	Datum
Kwajalein	0.094m	0.407m	MSL epoch 1983-2001
Ebeye	-0.004m	0.407m	EbeyeGVD19
Majuro	0.035m	0.433m	MGVD17

Table 1: MSL Mean and StdDev WRT local datum

The results in Table 1 and the comparison graphs indicate that tide observations in this region can be used to transfer datums from one area to another, as long as they have similar tidal signals. This is commonly used in hydrographic surveys where chart datum is transferred from a known location to an unknown location. All that is required to achieve such a transfer is a primary gauge operating at the known datum location and a temporary gauge operating simultaneously at the location where the datum is unknown. Transfer utilising this method usually only requires 2 to 7 days of simultaneous observations, however longer durations provide improved results.

This method of water level transfer can also be used to establish datums (MSL, high water, low water) at the remote locations from sites with a much longer tidal record (such as Majuro and Kwajalein). This method can provide a consistent geodetic reference, and it reduces the need for long leveling campaigns, which in many cases is not possible.

5 Conclusion and Recommendations

Conclusion

A new geoid was created for the entire Kwajalein Atoll, including Ebeye, by vertically adjusting the EGM08 geoid. The new geoid was adjusted to MSL as defined by the NOAA 2019 survey and 7 months of tide gauge observations at Ebeye.

The final geoid provides a model to translate between ellipsoid heights, referenced to ITRF2014 epoch 2019, and orthometric heights, referenced to MSL as defined by the 2019 survey 1.468m above tide gauge zero and 1.482m below BMWHRF. This datum is referred to as “EbeyeGVD19”.

No other orthometric elevations were available at the time of this report; therefore, the Geoid should be validated in the field when possible.

This Report should be read in conjunction with ‘*IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*’ and ‘*IIC Majuro Geoid: Development of a Geoid for Majuro Atoll, Republic of Marshall Islands dated 26 May 2020*’.

Recommendations

It is recommended that:

- GNSS observations are conducted on a BM for Kwajalein to establish the geoid ellipsoid link.
- Levelling be undertaken between some of the BMs and control points in Ebeye to verify the geoid.
- Consider the possibility of transferring datums via water level transfer. This may be an efficient way to transfer datums between various locations within an atoll, or even from one atoll to another.